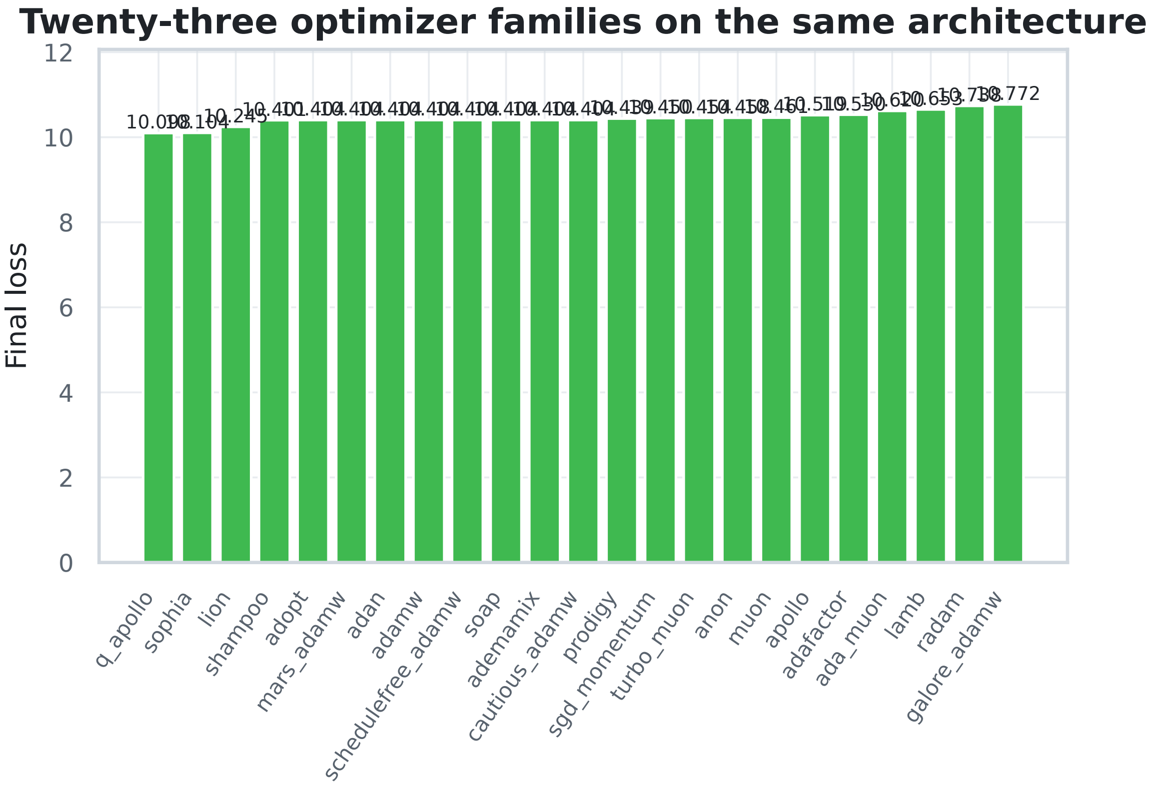
Twenty-three optimizer families on the same architecture
Final loss
0
2
4
6
8
10
12
10.098
10.104
10.245
10.401
10.404
10.404
10.404
10.404
10.404
10.404
10.404
10.404
10.430
10.450
10.454
10.458
10.461
10.519
10.530
10.620
10.653
10.738
10.772
q_apollo
sophia
lion
shampoo
adopt
mars_adamw
adan
adamw
schedulefree_adamw
soap
ademamix
cautious_adamw
prodigy
sgd_momentum
turbo_muon
anon
muon
apollo
adafactor
ada_muon
lamb
radam
galore_adamw